King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The parametric equations

$$x = 3\csc\theta - 1, \quad y = 4\cot\theta - 1$$

represent

(a) A parabola

(b) A hyperbola

(c) A pair of intersecting lines

(d) A line

(e) An ellipse

2. The area of the region that lies inside $r = 4\sin\theta$ and outside $r = 2$ is equal to

(a) $\int_{\pi/6}^{5\pi/6} \left(4\sin^2\theta - 1\right) d\theta$

(b) $4\int_{\pi/6}^{\pi/2} \left(4\sin^2\theta - 1\right) d\theta$

(c) $8\int_{\pi/6}^{5\pi/6} \left(1 - 4\sin^2\theta\right) d\theta$

(d) $8\int_{\pi/6}^{\pi/2} \left(4\sin^2\theta - 1\right) d\theta$

(e) $4\int_{\pi/6}^{\pi/2} \left(1 - 4\sin^2\theta\right) d\theta$

3. Consider the following statements about vectors. (I) If $\vec{u}$ and $\vec{v}$ have the same magnitude and direction, then $\vec{u}$ and $\vec{v}$ are equivalent. (II) If $\vec{u}$ and $\vec{v}$ are non-zero vectors sharing the same direction, then $\vec{v} = \|\vec{v}\|\,\vec{u}$. (III) If $a = b$, then $\|a\vec{i} + b\vec{j}\| = \sqrt{2}\,a$. (IV) If $\vec{u}$ and $\vec{v}$ have the same magnitude but opposite directions, then $\vec{u} + \vec{v} = \vec{0}$. Then

(a) II and IV are true

(b) I and III are true

(c) I and II are true

(d) Only I is true

(e) I and IV are true

4. If $\vec{w}$ is a vector with length 6 and has the same direction as $\vec{v} = \langle 2, 0, \sqrt{12} \rangle$, then $\vec{w} =$

(a) $\langle 3, 0, 3\sqrt{3} \rangle$

(b) $\langle \sqrt{6}, 0, \sqrt{30} \rangle$

(c) $\langle 3\sqrt{3}, 0, 3 \rangle$

(d) $\langle 0, 3, 3\sqrt{3} \rangle$

(e) $\langle 12, 0, 12\sqrt{12} \rangle$

5. If $\vec{u} = \langle a, 1, 0 \rangle$, $\vec{v} = \langle 0, a, 1 \rangle$, $\vec{w} = \langle 1, 0, a \rangle$ are such that $\vec{u} \cdot (\vec{v} \times \vec{w}) = 9$, then $a =$

(a) 4

(b) 0

(c) 2

(d) 3

(e) 1

6. The slope of the tangent line to the polar curve $r = 3 \sin \theta$ at the point $\left(\frac{3}{2}, \frac{\pi}{6} \right)$ is equal to

(a) $\frac{1}{2}$

(b) $\frac{1}{\sqrt{3}}$

(c) $\frac{-2}{\sqrt{3}}$

(d) $-\sqrt{3}$

(e) $\sqrt{3}$

7. The area of the surface formed by revolving the right part of the cardioid $r = 1 + \sin \theta$, $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ about the line $\theta = \frac{\pi}{2}$ is

(a) $\frac{4\pi}{5}$

(b) $\frac{8\pi}{5}$

(c) $\frac{16\pi}{5}$

(d) $\frac{32\pi}{5}$

(e) $\frac{2\pi}{5}$

8. The area of the surface formed by revolving the curve represented by the equations

$$x = 2\cos t, \quad y = 2\sin t, \quad 0 \le t \le \frac{\pi}{3}$$

about the x-axis is

(a) 4π

(b) π

(c) 2π

(d) 5π

(e) 6π

9. The arc length of the curve represented by the parametric equations

$$x = e^{-t}\cos t, \quad y = e^{-t}\sin t, \quad 0 \le t \le \frac{\pi}{2}$$

is

(a) $\sqrt{3}\left(1 + e^{\pi/2}\right)$

(b) $3\left(1 + e^{\pi/2}\right)$

(c) $2 - 2e^{-\pi/2}$

(d) $2\left(1 + e^{-\pi/2}\right)$

(e) $\sqrt{2}\left(1 - e^{-\pi/2}\right)$

10. The equation of the sphere that has a diameter with end points $(5, 0, -1)$ and $(1, -4, 1)$ is

(a) $(x-3) + y^2 + z^2 = 9$

(b) $(x-3)^2 + (y-2)^2 + z^2 = 9$

(c) $(x+3)^2 + (y+2)^2 + z^2 = 9$

(d) $(x-3)^2 + (y+2)^2 + z^2 = 9$

(e) $(x-3) + y^2 + (z-1)^2 = 9$

11. The polar coordinates (r, θ) corresponding to the rectangular coordinates $(5, -5\sqrt{3})$ with $r < 0$ and $0 \leq \theta < 2\pi$ is

(a) $\left(-10, \dfrac{5\pi}{3}\right)$

(b) $\left(-5, \dfrac{2\pi}{3}\right)$

(c) $\left(-10, \dfrac{2\pi}{3}\right)$

(d) $\left(-5, \dfrac{\pi}{3}\right)$

(e) $\left(-10, \dfrac{\pi}{3}\right)$

12. Let $\vec{u}$ and $\vec{v}$ be two vectors such that $\vec{u} \cdot \vec{v} = -6$ and $\vec{v} = \langle 1, 2, k \rangle$. If $\text{proj}_{\vec{v}}\, \vec{u} = -\vec{v}$ and $k > 0$, then $k =$

(a) $\dfrac{1}{2}$

(b) $\dfrac{1}{3}$

(c) 1

(d) $\sqrt{3}$

(e) $\sqrt{2}$

13. The equation of the tangent line to the curve represented by the parametric equations $x = t^4 + 2,\ y = t^3 + t$ at the point $(3, -2)$ is

(a) $2y + 3x = 0$

(b) $y = 1 - x$

(c) $y = 5 + x$

(d) $2y = -x + 4$

(e) $y = 7 + 2x$

14. Suppose that the three vectors $\vec{u} = 3\vec{i} + c\vec{j} + \vec{k}$, $\vec{v} = 2\vec{j} - 2\vec{k}$, and $\vec{w} = \langle 3, 1, 1 \rangle$ are coplanar. Then $c =$

(a) -1

(b) 0

(c) 1

(d) 2

(e) -2

15. Let $\vec{u} = \langle 1, 2, -1 \rangle$ and $\vec{v} = \langle -1, 1, k \rangle$. If $\vec{u}$ is orthogonal to $2\vec{u} - \vec{v}$, then $k =$

(a) -9

(b) -8

(c) -11

(d) 11

(e) 8

16. The angle between the vectors $\vec{u} = \langle -2, 0, 1 \rangle$ and $\vec{v} = \langle 4, 0, -2 \rangle$ is equal to

(a) 2π

(b) 3π

(c) $\dfrac{\pi}{2}$

(d) $\dfrac{\pi}{3}$

(e) π

17. Which of the following has the same graph as the graph of the parametric equations $x = 2t,\ y = 4t - 3$?

(a) $x = \theta^3,\ y = 2\theta^3 - 3$

(b) $x = \dfrac{2}{\theta},\ y = \dfrac{4}{\theta} - 3$

(c) $x = 2t^2,\ y = 4t^2 - 3$

(d) $x = 2\cos\theta,\ y = 4\cos\theta - 3$

(e) $x = e^t,\ y = 2e^t - 3$

18. The arc length of the polar curve $r = 2\cos\theta$, $0 \leq \theta \leq \dfrac{\pi}{2}$ is equal to

(a) $\dfrac{2\pi}{3}$

(b) $\dfrac{3\pi}{2}$

(c) π

(d) 2π

(e) 3π

19. A unit vector that is perpendicular to the graph of the function $f(x) = -x^2 + 5$ at the point $(1, 4)$ is

(a) $\left\langle \frac{1}{\sqrt{5}}, \frac{-2}{\sqrt{5}} \right\rangle$

(b) $\left\langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$

(c) $\left\langle -\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}} \right\rangle$

(d) $\left\langle \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right\rangle$

(e) $\left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$

20. The area of the parallelogram with vectors $\langle 1, 2, 0 \rangle$ and $\langle -1, -3, 4 \rangle$ as adjacent sides is equal to

(a) 7

(b) 6

(c) 9

(d) 10

(e) 8

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$\csc^2\theta - \cot^2\theta = 1 \Rightarrow \frac{(x+1)^2}{9} - \frac{(y+1)^2}{16} = 1$, a hyperbola.
2	**(b)**	Intersect at $\frac{\pi}{6}, \frac{5\pi}{6}$; $A = \frac{1}{2}\int(16\sin^2\theta - 4)d\theta = 4\int_{\pi/6}^{\pi/2}(4\sin^2\theta - 1)d\theta$.
3	**(e)**	I and IV are true; II needs unit $\vec{u}$, III needs $\lvert a\rvert$.
4	**(a)**	$\lVert\vec{v}\rVert = 4$, $\vec{w} = \frac{6}{4}\vec{v} = \langle 3, 0, 3\sqrt{3}\rangle$.
5	**(c)**	Determinant $= a^3 + 1 = 9$, so $a = 2$.
6	**(e)**	$\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta} = \frac{3\sqrt{3}/2}{3/2} = \sqrt{3}$.
7	**(d)**	$S = 2\sqrt{2}\pi\int_{-\pi/2}^{\pi/2}(1+\sin\theta)^{3/2}\cos\theta\,d\theta = \frac{32\pi}{5}$.
8	**(a)**	$ds = 2\,dt$, $S = \int_0^{\pi/3} 8\pi\sin t\,dt = 8\pi\cdot\frac{1}{2} = 4\pi$.
9	**(e)**	$x'^2 + y'^2 = 2e^{-2t}$, so $L = \sqrt{2}\int_0^{\pi/2} e^{-t}dt = \sqrt{2}(1 - e^{-\pi/2})$.
10	**(d)**	Centre $(3, -2, 0)$, diameter 6 so $r = 3$: $(x-3)^2 + (y+2)^2 + z^2 = 9$.
11	**(c)**	$\lvert r\rvert = 10$; point is $(10, \frac{5\pi}{3}) = (-10, \frac{2\pi}{3})$.
12	**(c)**	$\frac{-6}{\lVert v\rVert^2} = -1 \Rightarrow \lVert v\rVert^2 = 6 = 5 + k^2$, so $k = 1$.
13	**(b)**	$t = -1$, slope $\frac{3t^2+1}{4t^3} = -1$, so $y + 2 = -(x - 3)$, i.e. $y = 1 - x$.
14	**(c)**	Triple product $= 6 - 6c = 0$, so $c = 1$.
15	**(c)**	$\vec{u}\cdot(2\vec{u} - \vec{v}) = 11 + k = 0$, so $k = -11$.
16	**(e)**	$\cos\theta = \frac{-10}{\sqrt{5}\cdot 2\sqrt{5}} = -1$, so $\theta = \pi$.
17	**(a)**	Given curve is the full line $y = 2x - 3$; only $x = \theta^3$ covers all real x.
18	**(c)**	$r^2 + r'^2 = 4$, so $L = \int_0^{\pi/2} 2\,d\theta = \pi$.
19	**(c)**	Tangent $\langle 1, -2\rangle$, normal $\pm\langle 2, 1\rangle/\sqrt{5}$; listed: $\langle -\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\rangle$.
20	**(c)**	Cross product $\langle 8, -4, -1\rangle$, length $\sqrt{81} = 9$.